Unit Joystick2

SKU:U024-V2













Description

Unit Joystick2 is a high-precision Hall effect joystick control unit, built with an STM32G030F6P6 main control chip and integrated firmware for joystick signal acquisition. The unit communicates via the I2C interface and supports three-axis control signal input, including analog input for the X/Y axes and digital button input for the Z axis. **Unit Joystick2** uses a Hall effect joystick, achieving high-precision control by detecting magnetic field changes, offering advantages such as contactless operation, wear resistance, high accuracy, and strong anti-interference capability, ensuring product stability and long service life. Additionally, the device is equipped with a WS2812 RGB LED for status indication and interactive display, and reserves a firmware upgrade interface for the STM32, making firmware updates convenient. This product is suitable for various application scenarios such as game controllers and robot control.

Features

- Hall effect joystick
- Three-axis input:
  - X/Y axis offset analog input
  - Z-axis button digital input
- Built-in STM32
- I2C communication
- RGB LED
- 2 x LEGO compatible holes

- Development Platform: Arduino, UiFlow

Includes

- 1 x Unit Joystick2
- 1 x HY2.0-4P Grove Cable (20cm)

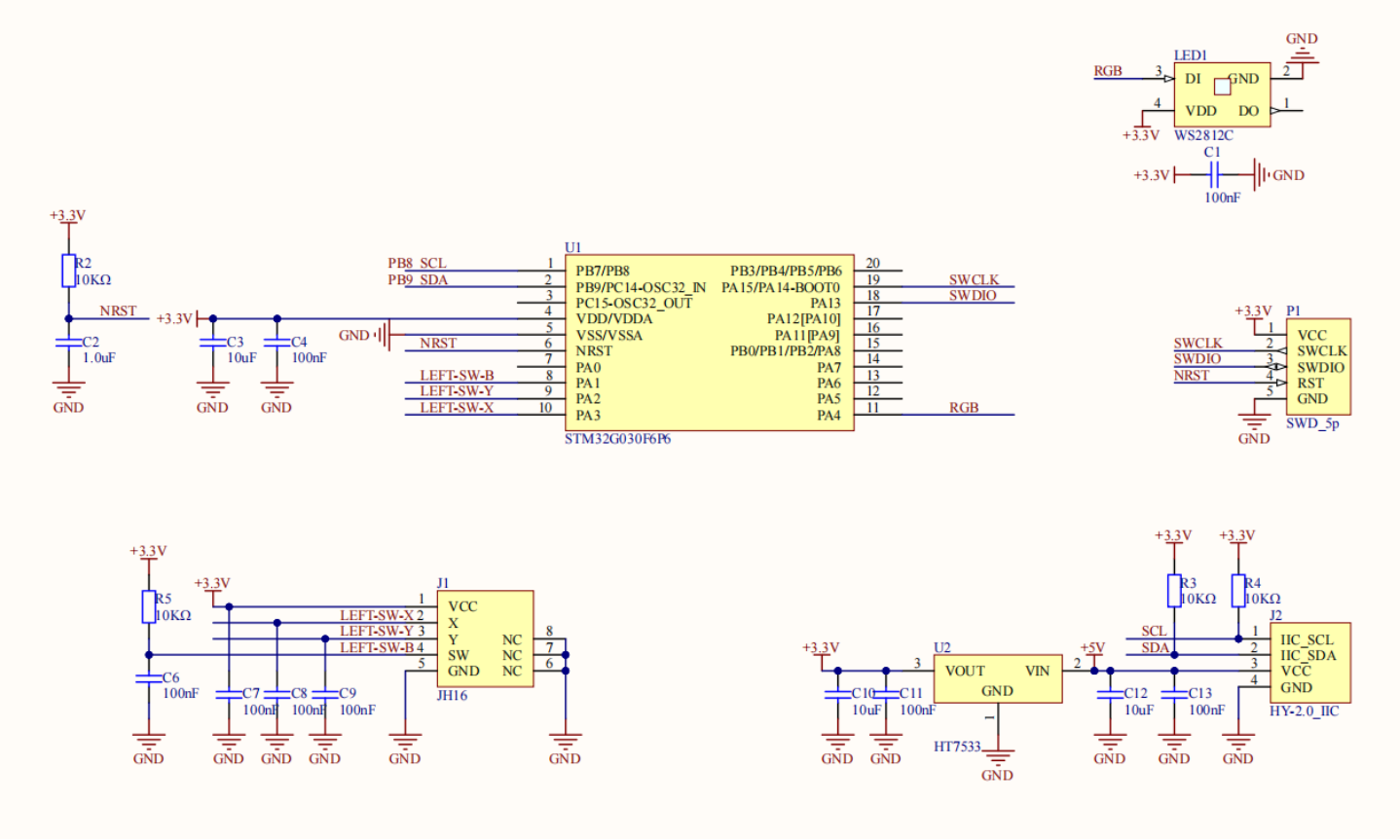
Applications

- Game controller
- Remote-controlled robot

Specifications

Specification	Parameter
MCU	STM32G030F6P6@Cortex-M0+, 64MHz main frequency
Flash	32KB
SRAM	8KB
Communication Interface	I2C Communication @0x63
X/Y axis output	16-bit ADC output (0-65535)
Z-axis button	0/1
RGB LED	1x WS2812C
Product Size	40.0 x 24.0 x 23.9mm
Product Weight	10.0g
Package Size	138.0 x 93.0 x 24.9mm
Gross Weight	15.7g

Schematics



PinMap

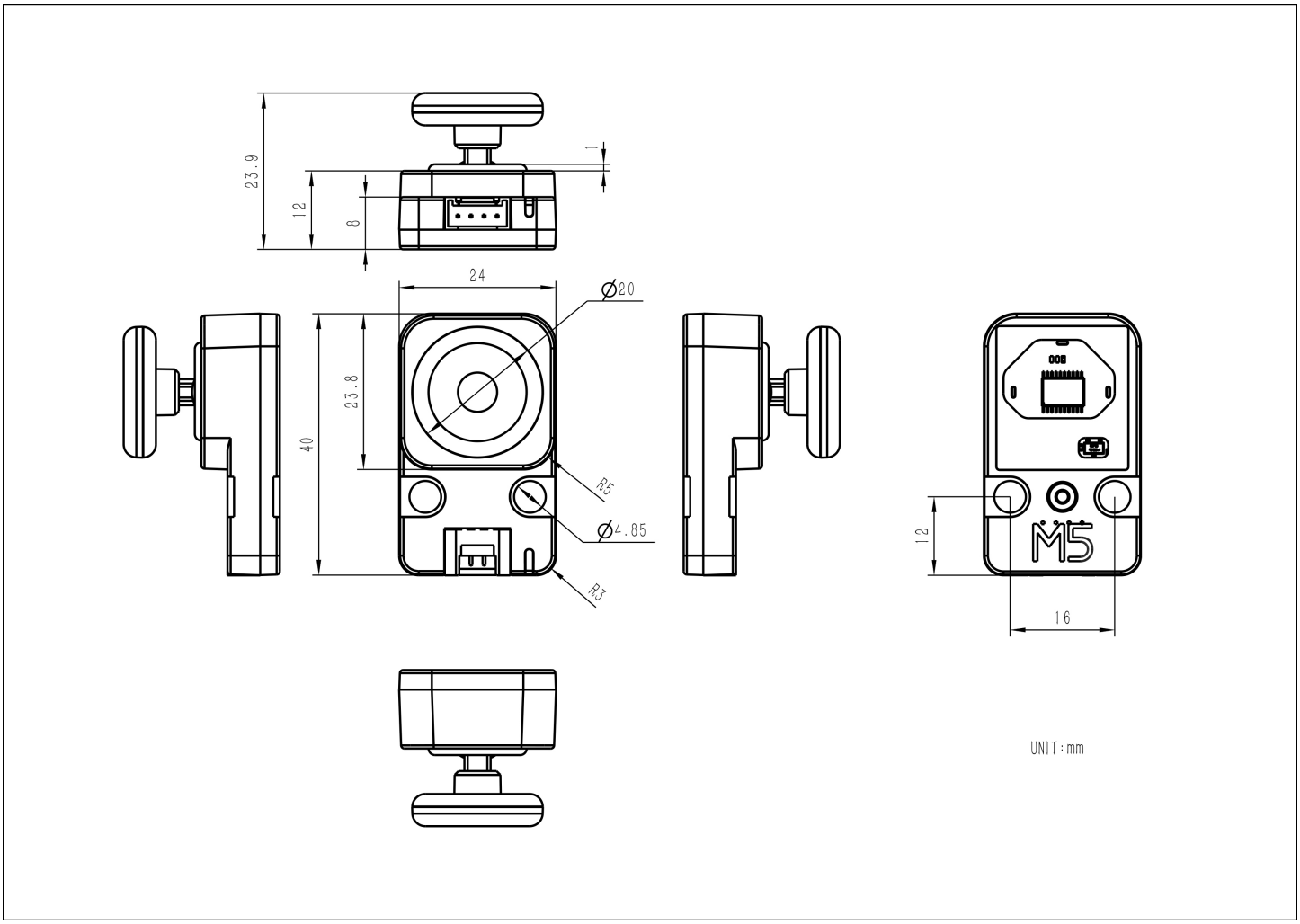
Unit Joystick2

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

STM32G030F6P6

STM32G030F6P6	PA1	PA2	PA3	PA4
X-Axis	LEFT-SW-X			
Y-Axis		LEFT-SW-Y		
Button			LEFT-SW-B	
WS2812C				RGB

Model Size



Structure

- Unit Joystick2 Structure Files

Softwares

Arduino

- Unit Joystick2 Arduino Library

UiFlow1

- Unit Joystick2 UiFlow1 Docs

UiFlow2

- Unit Joystick2 UiFlow2 Docs

Internal Firmware

- Unit Joystick2 Internal Firmware

Protocol

M5Stack Unit JoyStick2 I2C Protocol																		V1 (FW Version) 2024/1/10
REG MAP (Addr:0x63)			0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Raw Value	Joy1 ADC Value (16Bits)	0x00 R	Joy1 X ADC Value-L (0X00)	Joy1 X ADC Value-H (0X01)	Joy1 Y ADC Value-L (0X02)	Joy1 Y ADC Value-H (0X03)													ADC Value: ADC = (ADC Value-L + ADC Value-H * 256) 0~65535
	Joy1 ADC Value (8Bits)	0x10 R	Joy1 X ADC Value (0X10)	Joy1 Y ADC Value (0X11)															0~255
	Button	0x20 R	Button (0X20)																Button: 1, no press; 0, press
RGB	RGB	0x30 R/W	RGB-B (0X30)	RGB-G (0X31)	RGB-R (0X32)	NULL (0X33)													RGB value
Mapped Value	Joy1 ADC Min&Max&Center Value (16Bits)	0x40 R/W	Joy1 X ADC Negative Min Value-L (0X40)	Joy1 X ADC Negative Min Value-H (0X41)	Joy1 X ADC Negative Max Value-L (0X42)	Joy1 X ADC Negative Max Value-H (0X43)	Joy1 X ADC Positive Min Value-L (0X44)	Joy1 X ADC Positive Min Value-H (0X45)	Joy1 X ADC Positive Max Value-L (0X46)	Joy1 X ADC Positive Max Value-H (0X47)	Joy1 Y ADC Negative Min Value-L (0X48)	Joy1 Y ADC Negative Min Value-H (0X49)	Joy1 Y ADC Negative Max Value-L (0X4A)	Joy1 Y ADC Negative Max Value-H (0X4B)	Joy1 Y ADC Positive Min Value-L (0X4C)	Joy1 Y ADC Positive Min Value-H (0X4D)	Joy1 Y ADC Positive Max Value-L (0X4E)	Joy1 Y ADC Positive Max Value-H (0X4F)	Joy1 X ADC Negative Min Value: The minimum ADC value on the negative half-axis of X. Joy1 X ADC Negative Max Value: The maximum ADC value on the negative half-axis of X. Joy1 X ADC Positive Min Value: The minimum ADC value on the positive half-axis of X. Joy1 X ADC Positive Max Value: The maximum ADC value on the positive half-axis of X. Joy1 Y ADC Negative Min Value: The minimum ADC value on the negative half-axis of Y. Joy1 Y ADC Negative Max Value: The maximum ADC value on the negative half-axis of Y. Joy1 Y ADC Positive Min Value: The minimum ADC value on the positive half-axis of Y. Joy1 Y ADC Positive Max Value: The maximum ADC value on the positive half-axis of Y.
	Int12 Value (with offset)	0x50 R	X-L (0X50)	X-H (0X51)	Y-L (0X52)	Y-H (0X53)													Int12 Value: -4095~4095
	Int8 Value (with offset)	0x60 R	X-L (0X60)	Y-L (0X61)															Int8 Value: -127~127
System	Bootloader Version	0xF0 R													Version (0XFC)				Version: Bootloader version number
	Firmware Version	0xF0 R															Version (0XFE)		Version: firmware version number
	I2C Address	0xF0 R/W																Address (0XFF)	Address: I2C Address

Video

- Unit Joystick2 product introduction and case demonstration

U024-V2 JoyStick2 Unit 视频.mp4

- UiFlow2 Unit Joystick2

Product Comparison

Product Compare		
	Unit Joystick v1.1	Unit Joystick2
MCU	MEGA8A	STM32G030F6P6
RGB	/	WS2812C
X/Y axis offset output value	8 bits (0-255)	16 bits (0-65535)
Communication Mode	I2C (0x52)	I2C (0x63)
manufacturing technique	Carbon film joystick	Hall effect joystick